

# Student's Declaration

I, DIAR BEGISBAYEV, certify that the work embodied in this Master's thesis, carried out by me under the supervision of AIVAR SAKHIPOV for the period of September 2024 to June 2026 at Astana IT University, has not been submitted for the award of any other degree or diploma except where due acknowledgement has been made in the text.

I further declare that the material obtained from other sources has been duly acknowledged in the thesis, and that the research was conducted in accordance with the ethical guidelines of Astana IT University.

SIGNED: .....  ..... DATE: ..... 03.06.2026 .....